TECHNICAL REPORT: CYBERSECURITY PENTESTING LAB SETUP

Course: 6-Month Cybersecurity and Ethical Hacking Internship

Module: Module 1 – Cybersecurity Foundations, Networking & Ethical Hacking Lab

Project Title: Mini Project 1: Build Your Own Cybersecurity Lab

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Date: October 19, 2026

1. Executive Summary

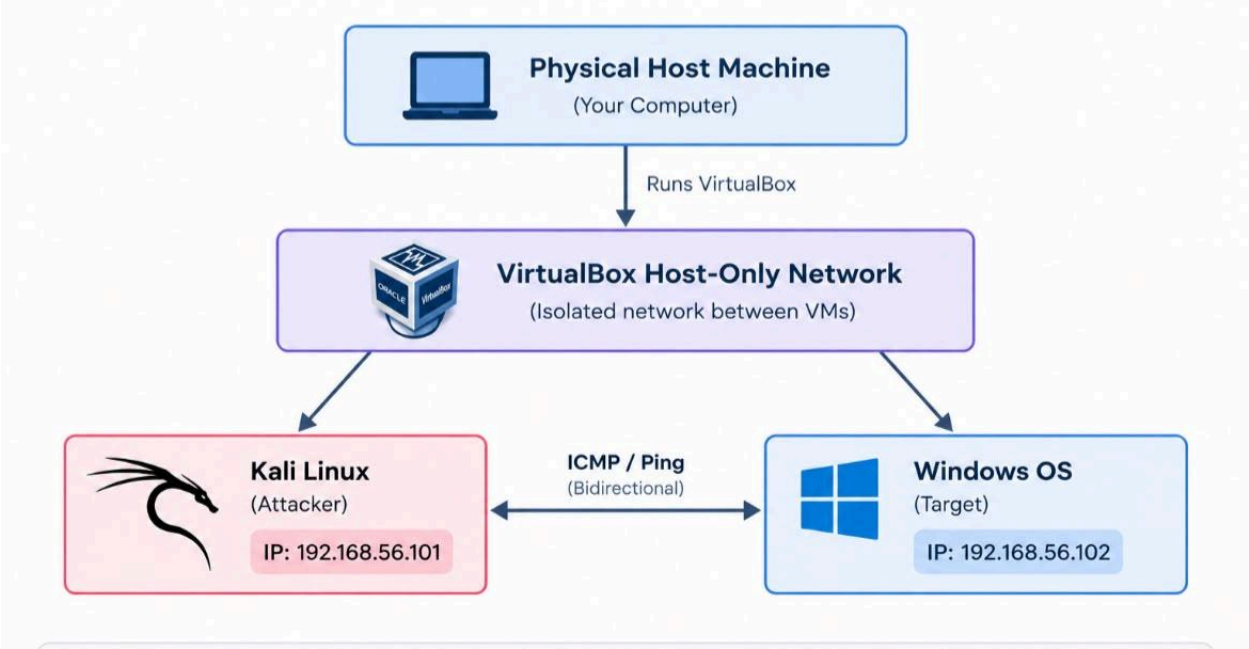
This technical report details the implementation, architecture, and validation of an isolated, dual-virtual machine (VM) cybersecurity lab environment. Designed specifically for penetration testing, packet analysis, and security

evaluation, this lab isolates attacking and target systems within a controlled environment to ensure host isolation and zero network spillover.

The lab consists of an attacker system (**Kali Linux**) and a target system (**Microsoft Windows**) connected via a dual-network adapter topology combining internet access (NAT) with isolated inter-system communication (Host-Only / Internal Network).

2. Lab Network Architecture

The environment relies on a segmented networking topology configured in Oracle VM VirtualBox:



Key Network Characteristics:

- **Primary Interface (NAT):** Provides outbound internet access for updating packages, tools, and repositories safely.

- **Secondary Interface (Host-Only / Internal Network):** Creates an isolated layer strictly dedicated to security testing, vulnerability scanning, and packet sniffing.

3. System Specifications & Configuration

Parameter	Attacking System (Kali Linux)	Target System (Windows OS)
Role	Attack Platform & Reconnaissance	Target / Victim Workstation
Operating System	Kali Linux 2026.2 Debian (64-bit)	Windows 10 (64-bit)
RAM Allocated	4096 MB (4 GB)	3082 MB (3 GB)
Processor Cores	2 vCPUs	2 vCPUs

Storage	30 GB (Dynamically Allocated)	40 GB (Dynamically Allocated)
Network Adapter 1	NAT	NAT
Network Adapter 2	Host-Only (vboxnet0)	Host-Only (vboxnet0)
Assigned IP	192.168.56.101	192.168.56.102

4. Step-by-Step Implementation

Step 1: Hypervisor & Virtual Machine Deployment

1. Installed **Oracle VM VirtualBox** on the host operating system.

2. Imported/Installed **Kali Linux** into VirtualBox with optimal memory and CPU allocation.

3. Created a new Virtual Machine for **Windows OS** using the installation ISO and allocated necessary system resources.

Step 2: Dual Network Adapter Setup

1. Opened VirtualBox **Settings > Network** for Kali Linux:
    - Configured **Adapter 1** as NAT.
    - Configured **Adapter 2** as Host-Only Adapter (Name: VirtualBox Host-Only Ethernet Adapter).
2. Repeated the exact dual-adapter configuration for the Windows Virtual Machine.

Step 3: Windows Firewall Customization (ICMP Echo Request)

To enable connectivity verification between systems:

1. Booted the Windows Target Machine.
2. Navigated to **Windows Defender Firewall with Advanced Security**.
3. Enabled the inbound rule: File and Printer Sharing (Echo Request - ICMPv4-In).

5. Verification & Testing Screenshots

Verification 1: Concurrent Execution in Hypervisor

Description: Both Kali Linux (Attacker) and Windows (Target) virtual machines actively running simultaneously within Oracle VM VirtualBox manager.

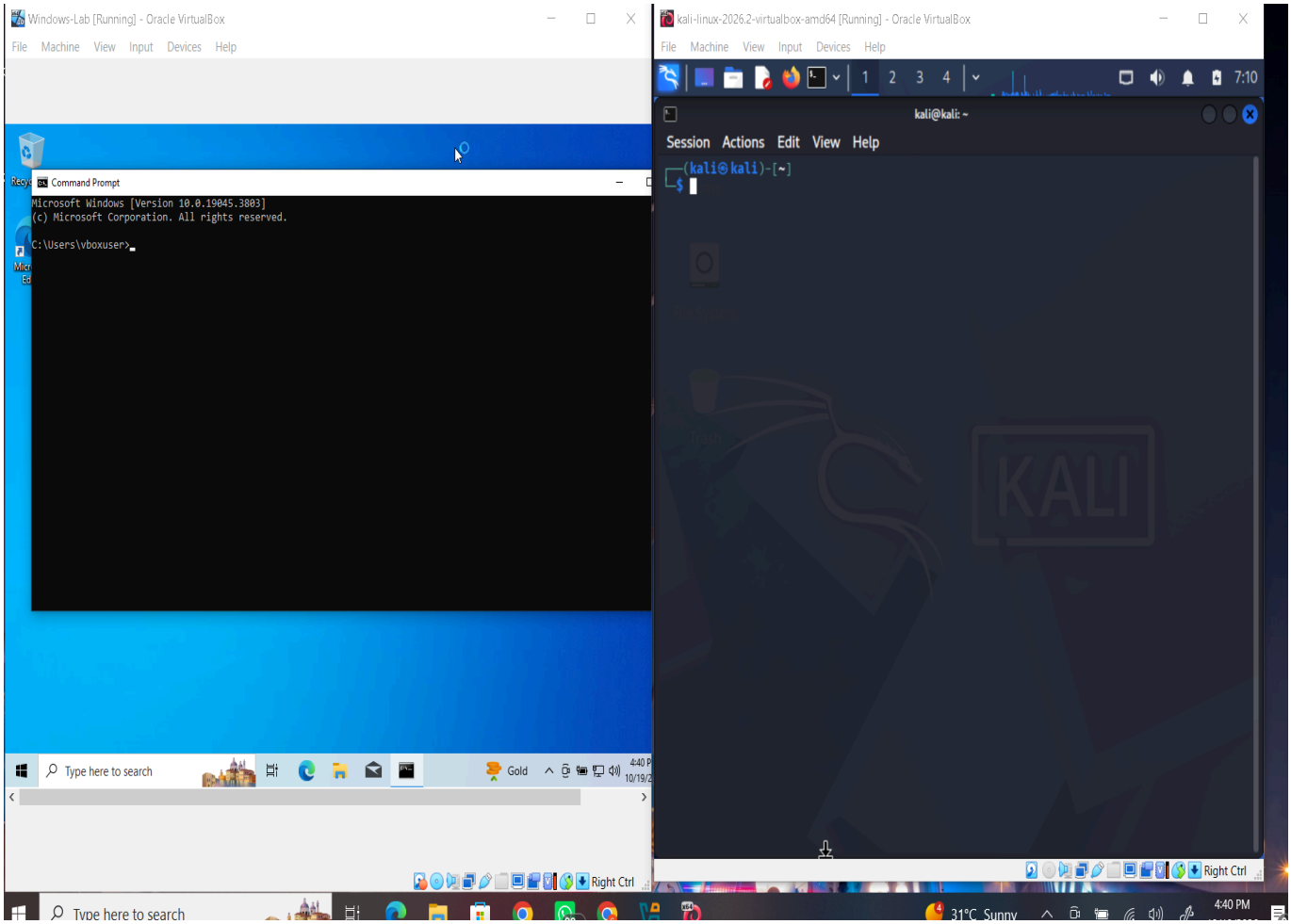


Figure 1.0: Active Execution of Attacker and Target Virtual Machines

Verification 2: IP Configuration & Interface Validation

Description: Terminal and Command Prompt outputs displaying the network interface configurations and assigned IP addresses for both machines.

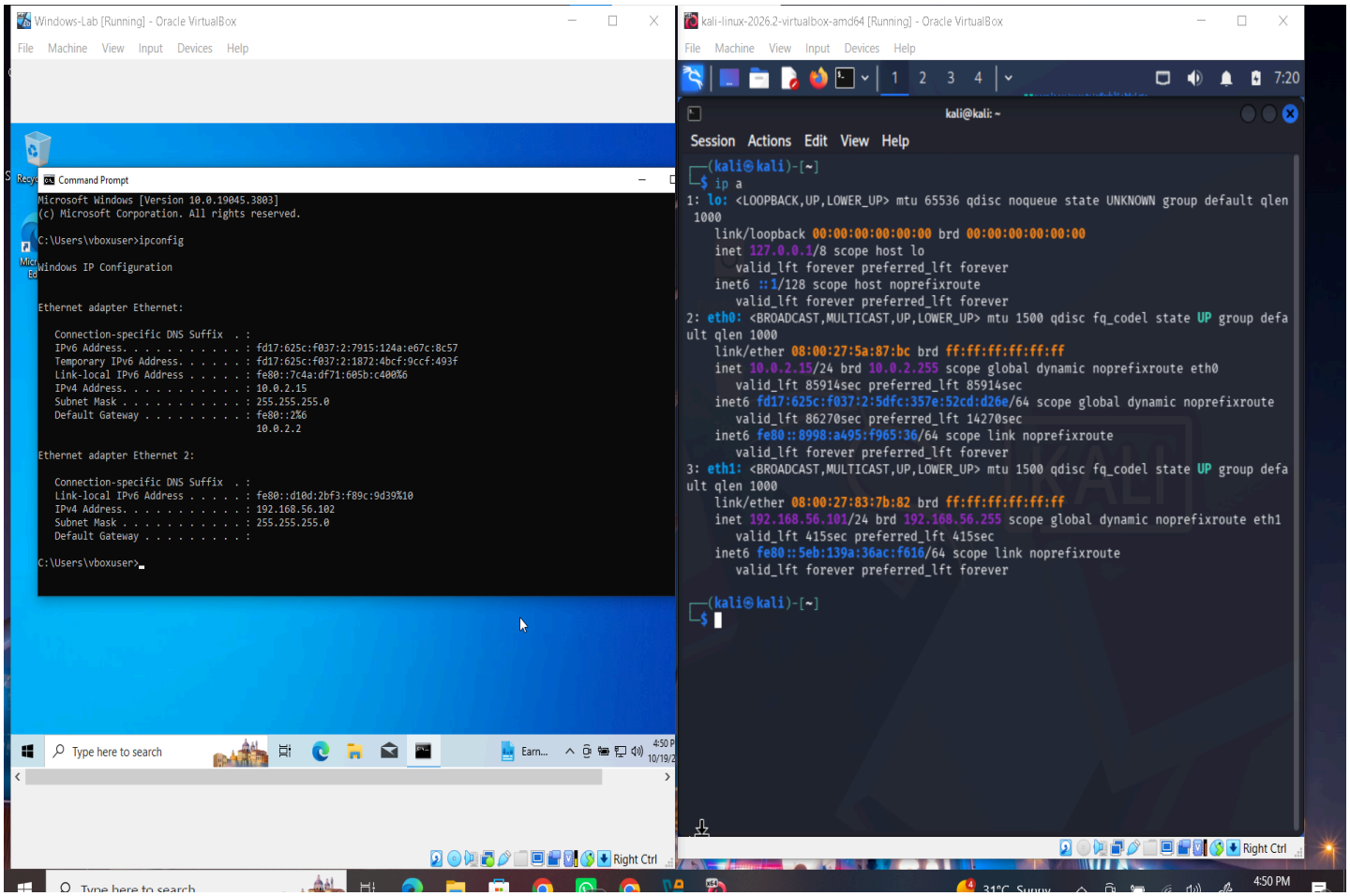


Figure 2.0: Network Interface Verification on Kali Linux and Windows

Verification 3: Inter-VM Connectivity Verification (Ping Test)

Description: Demonstration of ICMP connectivity between Kali Linux and the Windows target over the isolated Host-Only network layer.

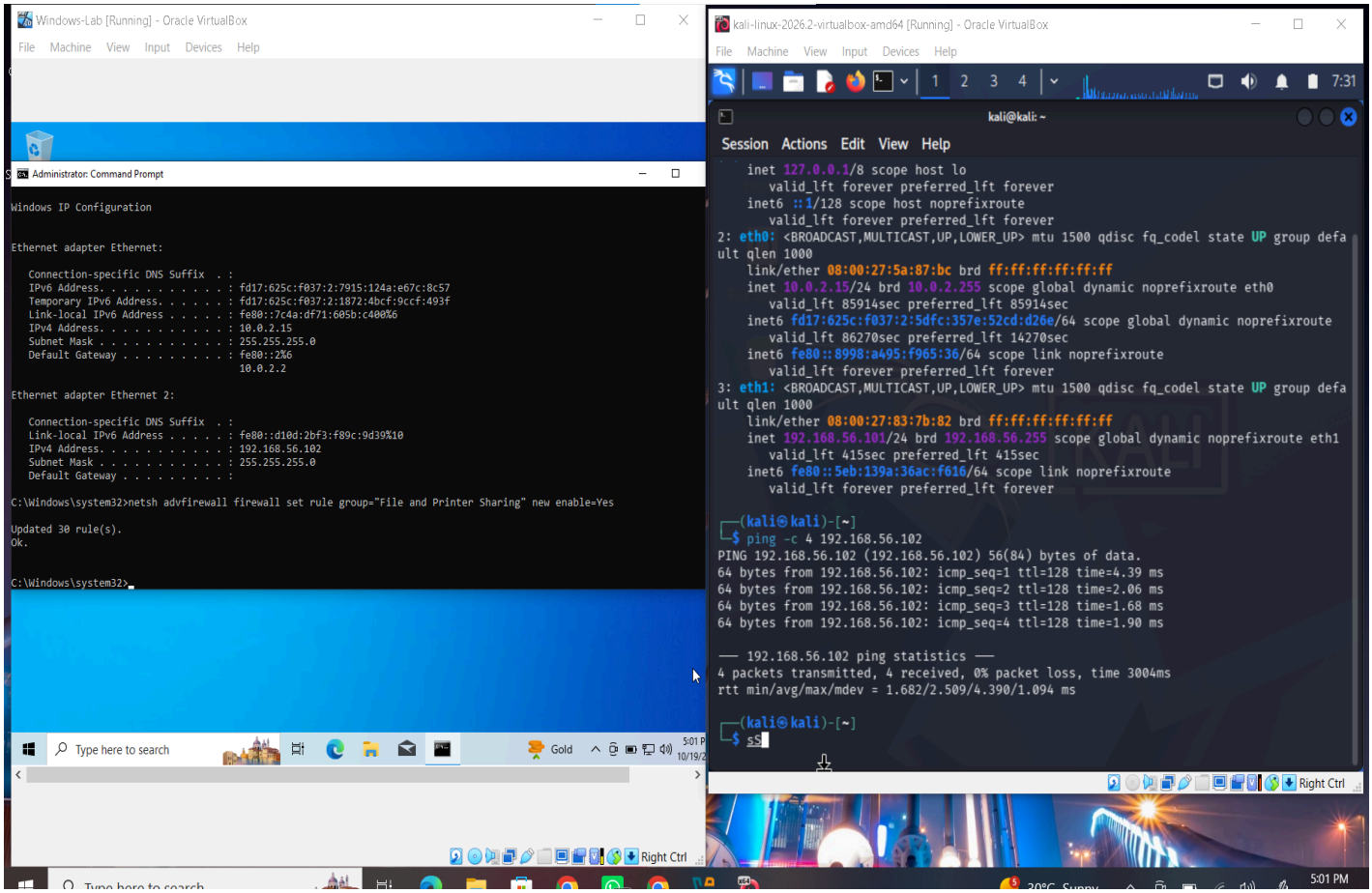


Figure 3.0: Successful ICMP Echo Response from Target Machine

6. Security & Isolation Controls

To ensure absolute safety during penetration testing procedures, the following security controls have been enforced:

1. **Host Isolation:** All traffic intended for penetration testing is routed strictly through the isolated Host-Only adapter.

2. **Promiscuous Mode Control:** Network interface promiscuous mode is restricted to virtual interfaces to avoid capturing real-world host traffic.

3. **Snapshot Baseline:** Initial clean system state snapshots were taken for both Kali Linux and Windows immediately after setup, enabling quick restoration if system files are modified during lab experiments.

7. Conclusion

The cybersecurity lab has been deployed, configured, and verified successfully. With working inter-system communication established over an isolated subnet, the environment is ready for network scanning, packet analysis, and ethical hacking experiments required in subsequent modules.